Modular Operator Discovery TSP Report

Overview

- Condition count: 7.
- Best final transfer gap: phase7_modular_operator_compression_pressure.
- Best held-out TSPLIB gap: phase7_full_solver_evolution.
- Surviving modular candidates: none.

Run Metadata

- run_name: run_20260520_021041_h
- started_at_local: 2026-05-20 02:10:41
- finished_at_local: 2026-05-20 02:41:58
- duration_hhmm: 00:31
- duration_seconds: 1876.298
- seed_offset: 7000
- replicate_label: h
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Mode	Replay	Selection	Final TSPLIB Gap	Final Family Gap	Final Transfer Gap	Mean Novelty	Mean Complexity	Surviving Candidate	Transplant Delta	Pareto Runtime Inflation
phase7_baseline_heuristic_only	baseline_only	none	score_only	0.235058	0.054968	0.121317	0.0	0.0	False	0.0	0.0
phase7_full_solver_evolution	full_solver	none	score_only	0.213387	0.064916	0.159398	0.845032	0.56	False	0.0	0.0
phase7_modular_operator_evolution	modular_operator	none	score_only	0.238552	0.065041	0.128966	0.0	0.445	False	0.024412	0.894435
phase7_modular_operator_random_replay	modular_operator	random	score_only	0.235058	0.054968	0.121317	0.785756	0.44	False	0.004184	-4.3e-05
phase7_modular_operator_diversity_residual_replay	modular_operator	diversity_residual	score_only	0.235058	0.054968	0.121317	0.908703	0.44	False	0.003509	0.005081

Condition	Mode	Replay	Selection	Final TSPLIB Gap	Final Family Gap	Final Transfer Gap	Mean Novelty	Mean Complexity	Surviving Candidate	Transplant Delta	Pareto Runtime Inflation
phase7_modular_operator_compression_pressure	modular_operator	none	novelty_gate	0.235058	0.050827	0.118702	0.796782	0.44	False	0.008919	0.977422
phase7_modular_operator_pareto_selection	modular_operator	none	pareto	0.235058	0.054968	0.121317	0.855895	0.4375	False	0.001654	-0.001558

Condition Notes

phase7_baseline_heuristic_only

- Execution mode baseline_only on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.054968, combined 0.121317.
- Accepted novelty 0.0, complexity 0.0, and adaptation efficiency 0.0.
- Last-epoch runtime 0.0 ms and distance evaluations 0.0.
- Validation: surviving False, transplant delta 0.0, positive scaffolds 0, Pareto runtime inflation 0.0.

phase7_full_solver_evolution

- Execution mode full_solver on host scaffold whole_solver.
- Final gaps: TSPLIB 0.213387, family holdout 0.064916, combined 0.159398.
- Accepted novelty 0.845032, complexity 0.56, and adaptation efficiency -0.029472.
- Last-epoch runtime 0.0 ms and distance evaluations 0.0.
- Validation: surviving False, transplant delta 0.0, positive scaffolds 0, Pareto runtime inflation 0.0.

phase7_modular_operator_evolution

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.238552, family holdout 0.065041, combined 0.128966.
- Accepted novelty 0.0, complexity 0.445, and adaptation efficiency 0.0.
- Last-epoch runtime 157.06945 ms and distance evaluations 6899.75.
- Validation: surviving False, transplant delta 0.024412, positive scaffolds 1, Pareto runtime inflation 0.894435.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 13, "complexity_score": 0.46, "distance_eval_inflation": 1.541079, "gap_delta": 0.004671, "runtime_inflation": 0.894435, "same_gap_faster": false}

phase7_modular_operator_random_replay

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.054968, combined 0.121317.
- Accepted novelty 0.785756, complexity 0.44, and adaptation efficiency 0.007573.
- Last-epoch runtime 77.9611 ms and distance evaluations 4863.0.

- Validation: surviving False, transplant delta 0.004184, positive scaffolds 0, Pareto runtime inflation -4.3e-05.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.0, "gap_delta": 0.0, "runtime_inflation": -4.3e-05, "same_gap_faster": true}

phase7_modular_operator_diversity_residual_replay

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.054968, combined 0.121317.
- Accepted novelty 0.908703, complexity 0.44, and adaptation efficiency 0.006548.
- Last-epoch runtime 296.338925 ms and distance evaluations 10000.75.
- Validation: surviving False, transplant delta 0.003509, positive scaffolds 0, Pareto runtime inflation 0.005081.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.0, "gap_delta": 0.0, "runtime_inflation": 0.005081, "same_gap_faster": false}

phase7_modular_operator_compression_pressure

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.050827, combined 0.118702.
- Accepted novelty 0.796782, complexity 0.44, and adaptation efficiency 0.018218.
- Last-epoch runtime 72.066475 ms and distance evaluations 4475.5.
- Validation: surviving False, transplant delta 0.008919, positive scaffolds 1, Pareto runtime inflation 0.977422.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.97364, "gap_delta": -0.003304, "runtime_inflation": 0.977422, "same_gap_faster": false}

phase7_modular_operator_pareto_selection

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.054968, combined 0.121317.
- Accepted novelty 0.855895, complexity 0.4375, and adaptation efficiency 0.013905.
- Last-epoch runtime 187.374425 ms and distance evaluations 7429.75.
- Validation: surviving False, transplant delta 0.001654, positive scaffolds 0, Pareto runtime inflation -0.001558.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 11, "complexity_score": 0.42, "distance_eval_inflation": 0.0, "gap_delta": 0.0, "runtime_inflation": -0.001558, "same_gap_faster": true}

Judge Appendix

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TSP Modular-Operator Discovery Suite Review

Key Evaluation Metrics (prioritized)

- **Held-out TSPLIB Gap** (transfer test)
- **Family Holdout Gap**
- **Operator Survival through Transplant & Ablation**

- **Evidence of Reusable Operator Structure**
- **Pareto Tradeoffs (gap vs runtime)**

Overview of Conditions

Condition Name	Transfer Gap	TSPLIB Gap	Family Gap	Transplant Mean Gap Delta	Ablation Support	Surviving Candidate	Operator Type	Comments Summary
phase7_baseline_heuristic_only	0.1213	0.2351	0.0550	0.0000	N/A	No	n/a	Baseline heuristic
phase7_full_solver_evolution	0.1594	0.2134	0.0649	0.0000	N/A	No	n/a	Full solver evolution, no operator survival
phase7_modular_operator_evolution	0.1290	0.2386	0.0650	0.0244	No	No	scaffold_selector	Some family gain but negative ablation; no transplant survival
phase7_modular_operator_random_replay	0.1213	0.2351	0.0550	0.0042	No	No	perturbation	No family gain, no ablation effect, no transplant survival
phase7_modular_operator_diversity_residual_replay	0.1213	0.2351	0.0550	0.0035	No	No	perturbation	Same as random_replay
phase7_modular_operator_compression_pressure	0.1187	0.2351	0.0508	0.0089	No	No	restart_controller	Some family help but no transplant survival; ablation does not confirm effect
phase7_modular_operator_pareto_selection	0.1213	0.2351	0.0550	0.0017	No	No	acceptance	No family gain, no ablation effect, no transplant survival

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Interpretation and Priority Analysis

1. **Held-Out TSPLIB Gap & Family Holdout Gap**

- None of the modular operators outperform baseline on held-out TSPLIB gap (transfer gap ≥ baseline 0.1213).
- Family holdout gap improvements are negligible or negative except for slight gain by "phase7_modular_operator_evolution" (+0.0650 vs baseline 0.0550) but this is not validated by transplant.

2. **Transplant Results**

- All modular operators fail clear transplant advantage:
- Mean transplant gap deltas are positive or near zero (worsening or no improvement).
- Most report "Does not transplant cleanly" failures.

- The best transplant positive scaffold counts are 0 or 1 with associated gap increases.

3. **Ablation Checks**

- For all modular operators, disabling the operator does **not hurt** performance (ablation gap delta ≤ 0).
- No ablation variant shows a meaningful improvement or operator necessity.
- This suggests no real, robust operator effect in modular components.

4. **Pareto Analysis**

- No operator improves gap without severe runtime inflation.
- The best compression_pressure operator has runtime inflation ~0.98 and minimal gap reduction.
- Others show zero or negative gap deltas with no runtime advantage.

5. **Operator Survival**

- None of the modular operators survive as candidates under transplant and ablation criteria.
- No evidence supports reusable operator structure surviving scaffold and family tests.
- Therefore, no claim for operator discovery or general reusable structure is warranted.

Conclusions

- **No modular operator passes the transplant and ablation robustness criteria.**
- **Held-out TSPLIB and family holdout gaps do not show consistent improvement from any operator.**
- **No operator demonstrates robust reusable structure; perturbation, acceptance, or restart operators do not survive validation.**
- **Given runtime inflation and lack of ablation support, operators fail to justify claim beyond baseline heuristics or full solver.**
- **Hence, no algorithm discovery or reusable operator claim is supported by data.**

Notable Observations

- The **"phase7_modular_operator_compression_pressure"** condition shows marginal family holdout gain in "two_cluster_bottleneck_tsp" (-0.024425 gap delta), but this does not transfer or survive transplant.
- Scaffold selector operator ("phase7_modular_operator_evolution") shows some family gain on "grid_like_tsp" but suffers from worse performance on held-out families and no ablation benefit.
- The baseline heuristic and full solver evolution conditions have higher training gaps but do not produce reusable operators.

Summary

No modular operator candidate demonstrates robust evidence of reusable operator structure or transfer performance improvements. The held-out TSPLIB and family results, combined with transplant and ablation checks, indicate failure to discover effective modular operators in this suite. Full-solver evolution maintains better gaps but does not yield reusable modular components. Focus should remain on improving operator validation and transplant success before asserting discovery claims.